

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0714 CSA07_CO2_AT3_Debuggine & Optimization task

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Renee Vincent ASR

Annexure A

Debugging & Optimisation Task

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

Given

- **Network Address:** 192.168.1.0/24
- **Subnet Masks Used:** /26, /27, /28

1. Identify Network, Broadcast, and Host Range

Subnet	Network Address	Host Range	Broadcast Address	Usable Hosts
192.168.1.0/26	192.168.1.0	192.168.1.1 – 192.168.1.62	192.168.1.63	62
192.168.1.64/27	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95	30
192.168.1.96/28	192.168.1.96	192.168.1.97 – 192.168.1.110	192.168.1.111	14

2. Calculate Usable and Unused IP Addresses

Assume department requirements are:

- Department A → 50 hosts
- Department B → 20 hosts
- Department C → 10 hosts

Department	Allocated Subnet	Usable IPs	Hosts Required	Unused IPs
A	/26	62	50	12
B	/27	30	20	10
C	/28	14	10	4

3. Identify Inefficient Subnet Allocations

- Department A wastes **12** IP addresses.
- Department B wastes **10** IP addresses.
- Department C wastes **4** IP addresses.
- If any department grows beyond its allocated size, it will experience an IP shortage despite unused addresses existing in other subnets.

4. Debug the Addressing Plan

Possible subnet allocation mistakes:

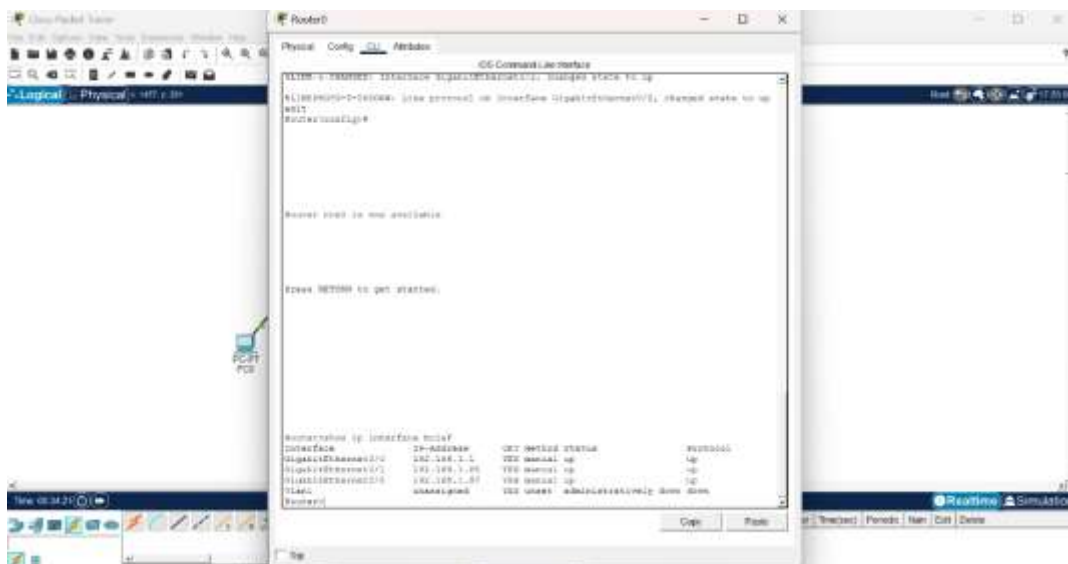
- Fixed-size subnet allocation instead of using **Variable Length Subnet Masking (VLSM)**.
- Some departments receive more addresses than needed, while others may not have enough for future growth.
- Poor utilization of the available /24 address space.

5. Optimized Subnetting Scheme Using VLSM

Allocate subnets from largest to smallest requirement.

Department	Hosts Required	Recommended Prefix	Network Address	Host Range	Broadcast Address
Department A	50	/26	192.168.1.0/26	192.168.1.1 – 192.168.1.62	192.168.1.63
Department B	20	/27	192.168.1.64/27	192.168.1.65 – 192.168.1.94	192.168.1.95
Department C	10	/28	192.168.1.96/28	192.168.1.97 – 192.168.1.110	192.168.1.111

The remaining addresses (192.168.1.112 – 192.168.1.255) can be reserved for future departments or expansion.



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation